Machine Learning

Support Vector Machines: Maximal Margin Classifier

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Support vector machines

- Support vector machines (SVM) are a very popular technique for classification in machine learning.
- Recall that:
  - LDA models the joint distribution of (Y, X);
  - logistic regression models the conditional distribution of $Y \mid X$.
- SVM is a geometric approach and makes no distributional assumption.
- SVM directly tries to find the separating hyperplane

$$x \in \mathbb{R}^p : \qquad \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = 0$$

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1. **Maximal Margin Classifier**: perfect linear separation if the classes are separable.
2. **Support Vector Classifier**: soft linear separation if the classes are not separable.
3. **Support Vector Machines**: soft non-linear separation (in the original feature space) if the classes are not separable.

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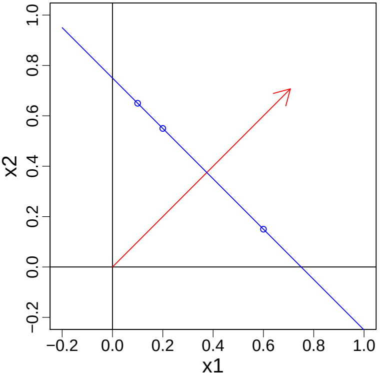
- SVM is a kernel method that relies on distances, so we will always assume that our predictors have been standardized to have mean 0 and variance 1.

Hyperplanes: construction

$$\text{Hyperplane:} \quad x \in \mathbb{R}^p : \quad \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = 0$$

- The vector $\beta = (\beta_1, \ldots, \beta_p)$ is called `normal vector`, it points in the direction that is `orthogonal` to the hyperplane.
- Without losing generality, it can be normalized to have norm one: $\|\beta\|^2 = \sum_{j=1}^{p} \beta_j^2 = 1$.
- Then: for a point $z \in \mathbb{R}^p$, $|\beta_0 + \beta_1 z_1 + \cdots + \beta_p z_p|$ is the `distance from the hyperplane`.



Separating hyperplanes

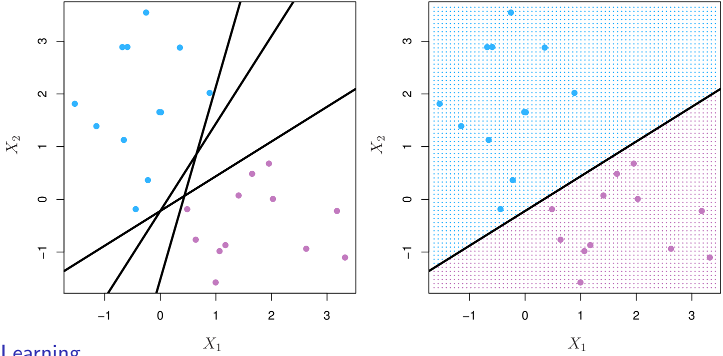
- We first look at two classes, and we will always identify them as $\mathcal{G} = \{-1, +1\}$.
- For data $(x_1, y_1), \ldots, (x_n, y_n)$, with $y_i \in \{-1, +1\}$, we say that a hyperplane is separating if

$$\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} > 0, \text{ if } y_i = +1,$$
$$\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} < 0, \text{ if } y_i = -1.$$

or equivalently, for all $i = 1, \ldots, n$

$$y_i(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}) > 0.$$

- If a separating hyperplane exists, there are infinitely many of them.



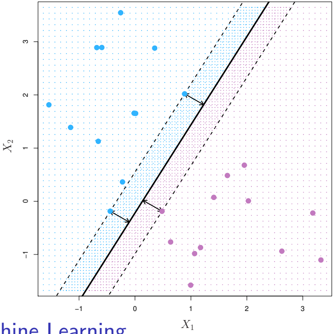
Maximal margin classifier

- If a separating hyperplane exists it is a `natural classifier` for a new test point $x_0 \in \mathbb{R}^p$:

$$f(x_0) = \beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p},$$

  and if $f(x_0) > 0$ is positive, we assign the class $+1$, and if $f(x_0) < 0$ we assign the class -1.
- The `magnitude` of $f(x_0)$ tells us how far it is from the hyperplane.
- The `margin` is the smallest distance of the training data from the hyperplane

$$M = \min_{i=1,\ldots,n} y_i(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}).$$



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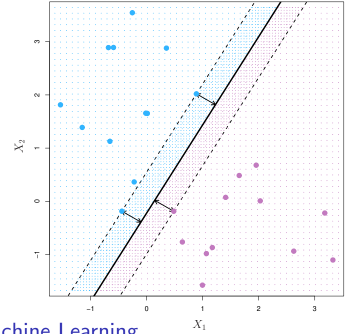
Maximal margin classifier: tries to make the `biggest gap` between the two classes.



Constrained optimization problem:

$$\underset{\beta_0,\ldots,\beta_p}{\text{maximize}}\ M$$

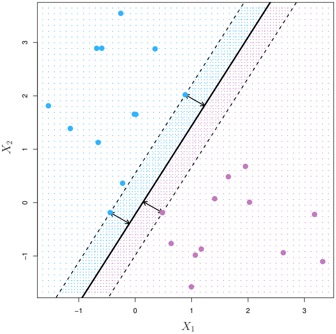
$$\text{subject to } \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}) \geq M,$$

$$\text{for all } i = 1, \ldots, n.$$

Comments and limitations

- This optimization problem is a `convex quadratic program` and can be solved efficiently.
- The observations (here three) determining the width of the margin are called `support vectors`. Moving the support vectors only slightly, the maximal margin hyperplane will move as well.
- Moving the other observations slightly, does not affect the hyperplane!



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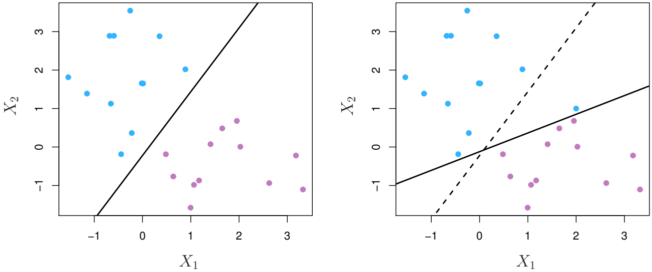
Limitations:

- The maximal margin hyperplane can be `unstable`, since it depends only on few support vectors.



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Limitations:

- The maximal margin hyperplane can be `unstable`, since it depends only on few support vectors.
- Most often, the data is `not separable`!